

1. Create an S3 bucket named 'music-uploads-demo' and upload a sample MP3 file to it using the AWS Management Console.

ANS :

- I created an S3 bucket named:

- music-uploads-demo

- ❖ Steps followed:

1. Logged in to the **AWS Management Console**.
2. Opened the **Amazon S3** service.
3. Clicked **Create bucket**.
4. Entered the bucket name:
5. music-uploads-demo
6. Selected the appropriate AWS Region.
7. Kept **Block all public access** enabled because the uploaded music file does not need to be publicly accessible.
8. Created the bucket.
9. Opened the bucket and selected **Upload**.
10. Selected a sample MP3 file from my computer.
11. Clicked **Upload**.

- The MP3 file was successfully uploaded to the S3 bucket.

2. Write an AWS Lambda function in Python called log_upload_event that logs the filename and upload timestamp whenever a new file is added to your 'music-uploads-demo' S3 bucket.

ANS :

- I created a Python AWS Lambda function named:

- log_upload_event

- The following code reads the S3 event and logs the uploaded filename and upload timestamp:

- import json type a code for it .

- The function extracts the S3 bucket name and uploaded filename from the S3 event notification and writes the information to CloudWatch Logs.

3. Configure an event trigger so that your `log_upload_event` Lambda function is automatically invoked every time a new file is uploaded to the 'music-uploads-demo' S3 bucket.

ANS :

- I configured an S3 event trigger so that the `log_upload_event` Lambda function runs automatically whenever a new object is uploaded.
- ❖ Steps followed:
 1. Opened the **S3** service.
 2. Opened the music-uploads-demo bucket.
 3. Went to **Properties**.
 4. Found **Event notifications**.
 5. Clicked **Create event notification**.
 6. Entered an event notification name such as:
 7. music-upload-trigger
 8. Selected the event type:
 9. All object create events
 10. Selected **Lambda Function** as the destination.
 11. Selected:
 12. `log_upload_event`
 13. Created the event notification.
- Now, whenever a new object is uploaded to the bucket, S3 sends an event to the Lambda function.

4. Simulate uploading a new file to your S3 bucket and verify that your Lambda function runs by checking the logs in AWS CloudWatch.

***Hint:** Look for the Lambda function's log group in CloudWatch to see the output for each upload event.*

ANS :

- I uploaded another sample file to the S3 bucket:
 - new-song.mp3
- The upload generated an S3 event, which automatically invoked the `log_upload_event` Lambda function.
- AWS Console → CloudWatch → Logs → Log groups
- Then I opened the Lambda function's log group:
- The logs contained information similar to:

- Bucket: music-uploads-demo
- File uploaded: new-song.mp3
- Upload timestamp: 2026-10-01T04:10:00+00:00
- This confirms that the S3 upload event successfully triggered the Lambda function.

5. Use ChatGPT or GitHub Copilot to suggest an additional event-driven use case for S3 and Lambda (other than logging uploads), and describe in 2-3 lines how you would implement it.

ANS :

- S3 can trigger a Lambda function whenever a new image is uploaded. The Lambda function can resize the image and save a smaller thumbnail to another S3 folder or bucket.
- ❖ Example implementation:
 - Image Upload → S3 → Lambda → Resize Image → S3 Thumbnail Folder
 - This can be used by applications such as food-delivery platforms to automatically create smaller versions of restaurant and food images after upload.